$$\rho_{aa'} = \sum_b \psi^*(a'b)\ \psi(ab). \qquad (7.31)$$

Consider this question: Can Bob, at his end, do anything to instantly change Alice's density matrix? Keep in mind that Bob can only do things that the laws of quantum mechanics allow. In particular, Bob's evolution, whatever causes it, must be unitary. In other words, it must be described by a unitary matrix

$$\mathbf{U}_{bb'}.$$

The matrix U represents whatever happens to Bob's system, whether or not Bob does an experiment. It acts on the wave function to produce a new wave function, which we'll call the "final" wave function:

$$\psi_{final}(ab) = \sum_{b'} \mathbf{U}_{bb'}\ \psi(ab').$$

We can also write the complex conjugate of this wave function:

$$\psi^*_{final}(a'b) = \sum_{b''} \psi^*(a'b'')\ \mathbf{U}^\dagger_{b''b}.$$

Notice that we added primes to some of the symbols to avoid mixing them up in the next step. Now, let's calculate Alice's new density matrix. We'll use Eq. 7.31, but we'll replace the original wave functions with the final ones:

$$\rho_{aa'} = \sum_{b,b',b''} \psi^*(a'b'')\ \mathbf{U}^\dagger_{b''b}\ \mathbf{U}_{bb'}\ \psi(ab').$$